

Problem

A Δ -connected source supplies power to a Y-connected load in a three-phase balanced system. Given that the line impedance is $2 + j1 \Omega$ per phase while the load impedance is $6 + j4 \Omega$ per phase, find the magnitude of the line voltage at the load. Assume the source phase voltage $\mathbf{V}_{ab} = 208 \angle 0^\circ \text{ V rms}$

Step-by-step solution

Step 1 of 2

Consider the following three phases balanced Δ -Y connection:

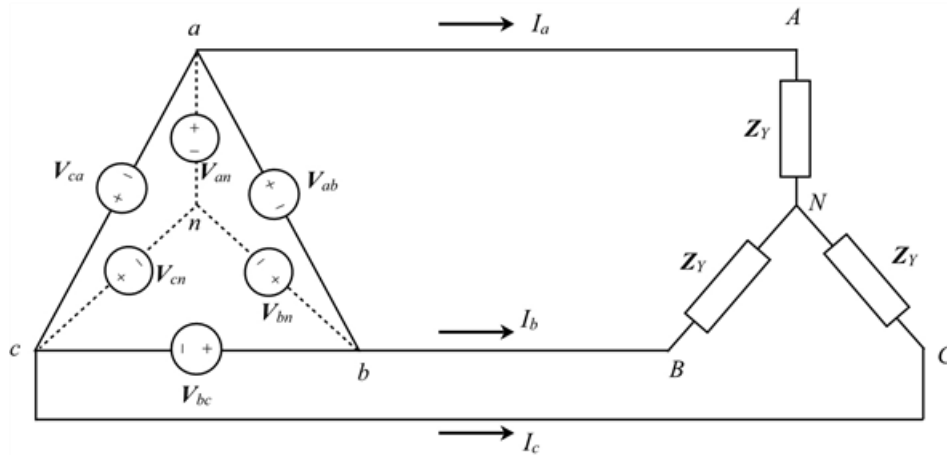


Figure 1

Consider the following data:

The line impedance per phase, $Z_L = 2 + j1 \Omega$

The load impedance per phase, $Z_l = 6 + j4 \Omega$

The source phase voltage, $\mathbf{V}_{ab} = 208 \angle 0^\circ \text{ V rms}$

In a balanced Δ -Y connection, the line and load impedances are in series.

Therefore,

$$\begin{aligned} Z_Y &= Z_L + Z_l \\ &= 2 + j1 + 6 + j4 \\ &= 8 + j5 \Omega \end{aligned}$$

The single phase equivalent circuit is shown in Figure 2.

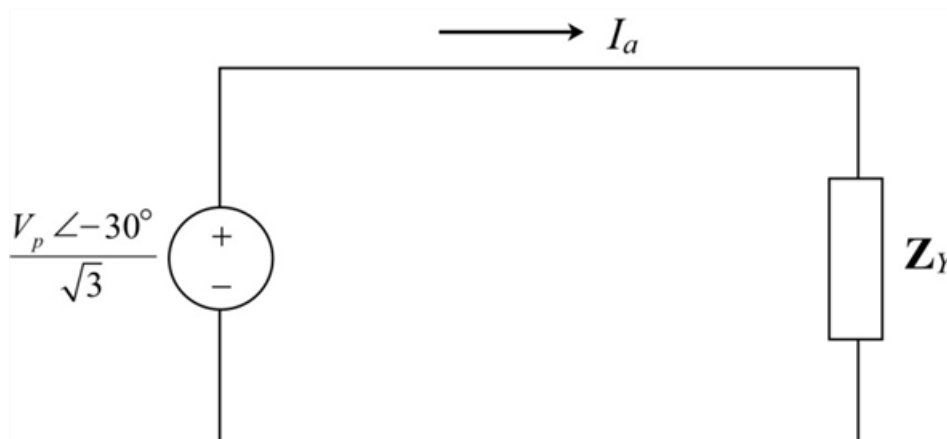


Figure 2

Calculate the current I_a .

$$\begin{aligned} I_a &= \frac{\frac{V_p \angle -30^\circ}{\sqrt{3}}}{Z_Y} \\ &= \frac{208 \angle -30^\circ}{(8 + j5)\sqrt{3}} \\ &= 12.73 \angle -62^\circ \text{ A} \end{aligned}$$

Now, calculate the load voltage, V_L .

$$V_L = (6 + j4)I_a \dots\dots (1)$$

Substitute $12.73 \angle -62^\circ \text{ A}$ for I_a in equation (1).

$$\begin{aligned} V_L &= (6 + j4)(12.73 \angle -62^\circ) \\ &= (7.211 \angle 33.7^\circ)(12.73 \angle -62^\circ) \\ &= 91.79 \angle -28.3^\circ \text{ V} \\ &= 80.75 - j43.48 \text{ V} \end{aligned}$$

Therefore, the magnitude of the line voltage at the load is 91.79 V .